

Moderating Multi-Agent Legal Role-Play with a Small Local Language Model

Design and rationale of the AI Law Lab moderator

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Abstract

Role-play between language-model agents is a promising instrument for legal education and research, but multi-agent exchanges fail in characteristic ways: agents repeat themselves, drift out of character, converge too early, or let one voice dominate. We describe the moderation logic of the AI Law Lab's role-play mode, which runs an 8-billion-parameter model on local hardware. The design separates judgement from rules: the model answers four structured questions about each turn, while code decides who speaks, when an impasse warrants intervention, and when a scene may end. Participants keep private negotiation ledgers and bottom lines that neither the moderator nor the other participants see. Each rule is grounded in prior work on LLM negotiation, multi-party turn-taking, multi-agent debate and mediation practice. A live 24-turn, three-party mediation illustrates the mechanisms and exposes two limitations: routing direct questions can override turn balance, and interventions can be addressed to a neutral participant.

1 Introduction

Legal education has long used simulated negotiations, mediations and hearings. Language-model agents make such simulations cheap to run at scale and, when every model call is logged, auditable, which makes them attractive both for teaching and for research on how models reason about legal disputes. The AI Law Lab runs these simulations on university hardware with an 8-billion-parameter model, so that unpublished or sensitive material never leaves campus.

Exchanges between language-model agents fail in recognizable ways. Negotiating agents without a plan re-propose the same deal round after round [1]; simulations can look varied yet end in impasse almost every time, far from human base rates [2]; without turn-taking structure, individual agents monopolize the floor or pose question after question [3]; agents abandon positions under peer pressure and collapse into premature agreement [4], [5]; and instruction-following drifts measurably within about eight rounds of dialog [6]. A small local model is, if anything, more exposed to these failures than the larger models studied in that work.

This note describes how the lab's role-play moderator addresses them. Its central design choice is to divide the work: the model supplies judgement about each turn in a constrained, structured form, and deterministic code applies the rules that decide who speaks, when to intervene and when a scene may end. Section 3 outlines the system and what each party may see, Section 4 specifies the moderation rules, Section 5 the participant-side mechanisms they rely on, and Section 6 the assessment. Section 7 walks through a live run and Section 8 sets out limitations.

2 Background

Conversation analysis describes how speakers allocate turns: the current speaker may select the next, most visibly by producing the first part of an adjacency pair such as a question, and otherwise participants select themselves [7]. Nonomura and Mori applied this ordering to multi-party dialog between LLM agents and found that letting the current speaker select the next, before any self-selection, reduced breakdowns such as one agent monopolizing the floor or asking question after question [3].

Principled negotiation separates positions from the interests behind them and asks each party to know its best alternative to a negotiated agreement (BATNA), which tells it when to walk away [8]. Mediators manage stalled talks with process tools, among them reality testing each side's assessment of its case, reframing and refocusing the discussion, and caucuses [9]. The moderator borrows these tools; the participant specification borrows the preparation.

3 System overview

A role-play is a LangGraph state machine over a shared transcript. In each cycle the moderator judges the latest turn and chooses the next speaker; the chosen participant first updates a private ledger and then speaks; the cycle repeats until the scene ends, after which an assessor evaluates the whole exchange. All model calls go to gemma4 (8 billion parameters, 4-bit quantized) served by Ollama on two NVIDIA DGX Spark units. Calls whose output code must read are constrained by a JSON schema. Every call, decision and intervention is written to an append-only trace.

Participants are specified by a role, an objective, and optional background, demeanor, behavioral tendencies and priorities, plus two private fields: a bottom line and confidential facts. Table 1 shows what each party can see. The separation protects validity: a moderator that saw bottom lines could steer the parties toward their zone of agreement, and a participant that saw another's ledger would no longer be negotiating.

Information	Speaking participant	Other participants	Moderator	Assessor
Scenario	yes	yes	yes	yes
Public transcript, including moderator statements	yes	yes	latest 8 entries	yes
Participants' names and roles	all	all	all	names only
Objective	own	own	all	no
Background, demeanor, tendencies, priorities	own	own	no	no
Bottom line and confidential facts	own	own	no	all
Private negotiation ledger	own	own	no	no
Moderator's private note (e.g. answer a question)	if addressed	no	writes it	no

Table 1. What each party sees during a run.

4 Moderation logic

4.1 Structured judgement

Before every turn after the first, the moderator is shown the scenario, each participant's role and objective, how often each has spoken, the seven preceding transcript entries shortened to 120 words each, and the latest entry in full. It must answer four questions in a fixed schema (Table 2). Reasoning is disabled, temperature is 0 and output is capped at 300 tokens: the task is classification, not analysis. Because `next_speaker` and `question_for` are constrained to the participants' identifiers, no free-text parsing is needed. The opening turn goes to the first listed participant without a model call.

Field	Question put to the model	Used by
<code>question_for</code>	Did the latest speaker put a direct question to, or demand an answer from, one specific other participant? (The person asked, never the asker, or "nobody".)	Rule 1, Section 4.2
<code>stalled</code>	Over the last several turns, are the parties restating the same positions without any new proposal, concession or question?	Interventions, Section 4.3
<code>concluded</code>	Have the parties clearly agreed on terms, or has someone clearly and finally walked away?	Ending, Section 4.4
<code>next_speaker</code>	Whose turn should come next for the exchange to be productive and fair?	Rules 2 to 4, Section 4.2

Table 2. The moderator's structured judgement. A fifth field, `reason`, is kept in the trace.

4.2 Speaker selection

Code turns the judgement into a choice with Algorithm 1. The previous speaker is never eligible. A direct question is answered first, following the adjacency-pair ordering of [7] that reduced breakdowns in [3]. The moderator's own suggestion stands unless it would put that participant two or more turns ahead of the quietest eligible participant, which guards against the monopolization that free self-selection produced in [3]. Ties among the quietest go to the next seat after the previous speaker. A participant chosen to answer a question also receives a private note to answer it before anything else.

```
input  participants in seating order; previous speaker p;
       turns spoken c(i); judgement: question_for q, next_speaker s

E <-  participants other than p
m <-  participant in E with the fewest turns (ties: next seat after p)

1  if q is in E                return q    answer the question
2  if s is not in E            return m    unusable suggestion
3  if c(s) - min{ c(i) : i in E } >= 2
                                return m    balance the floor
4  return s                    moderator's choice
```

Algorithm 1. Choosing the next speaker (`pick_speaker` in `roleplay_policy.py`).

4.3 Impasse interventions

When the judgement reports a stall, the moderator intervenes provided at least $\max(4, 2n)$ turns have passed since its previous intervention, where n is the number of participants. The cool-down gives the parties room to respond before it steps in again. Successive interventions rotate through three mediation techniques (Table 3). Each is addressed by name to the participant who speaks next, entered in the public transcript as a statement made in joint session, and repeated to that participant as a private note.

Technique	Statement to the next speaker	Basis
Reframe	"Before restating anything, say what you think the other side needs underneath its demand, and propose one option that could serve both sides."	Interests over positions [8]; reframing [9]
Narrow	"Pick the single open issue closest to agreement and make a concrete, specific proposal on that issue alone."	Refocusing the discussion [9]
Reality test	"Consider plainly what happens to your side if there is no agreement at all. Then either make a conditional offer or say clearly that you are prepared to walk away."	BATNA [8]; reality testing [9]

Table 3. Intervention techniques, used in rotation. Statements abridged.

4.4 Ending a scene

A scene ends at the configured maximum number of turns, or earlier when the judgement reports it concluded and every participant has spoken at least twice. The floor exists because multi-agent exchanges drift toward premature consensus and abandon positions under social pressure [4], [5]. A conclusion judged too early is written to the trace and the exchange continues.

5 Participant-side mechanisms

Before speaking, the selected participant updates a private negotiation ledger with seven fields: concessions made, concessions received, where things stand, its read on the others, open issues, arguments already made, and its next move. The update is a schema-constrained call with reasoning disabled that carries the previous ledger forward; a malformed update keeps the previous ledger. The design follows two findings: long-term planning notes kept negotiating agents from re-proposing similar deals [1], and in simulated negotiations an agent-authored ledger of concessions and open issues was the one intervention that moved outcomes off near-universal impasse, where extra output tokens, free-form reflection and reasoning modes did not [2]. The arguments-already-made field targets the repetition seen in the lab's earlier runs.

The reply prompt then presents, in order, what the participant has not yet responded to, its ledger, any note from the moderator, and a short character reminder restating its identity, objective, private bottom line and demeanor. The reminder comes last because adherence to initial instructions decays over long dialogs [6]. Replies use the model's reasoning mode with an output budget proportional to the word limit, and a reply exhausted by reasoning before producing any text is retried without it. Each participant's memory is scoped to that participant and stores every statement once, as what it heard since its last turn and what it said, compacting the oldest material into summaries beyond a token budget scaled to the turn length.

6 Assessment

After the scene an assessor receives the transcript together with the participants' bottom lines and confidential facts, which only it is shown, and answers five questions citing turn numbers: the outcome and its terms; which positions moved and why, and where the exchange repeated itself; whether anyone conceded past a bottom line or disclosed confidential information; any legal proposition that appears unsupported; and whether concessions and resistance were plausible for real counsel. A transcript above 60,000 tokens is split into segments of at most 24,000 tokens that are summarized concurrently, then assessed with the final six entries verbatim; a 100-turn scene at 1,000 words a turn would otherwise exceed the model's 128,000-token context.

7 Illustrative run

Table 4 summarizes a live run from September 2026. Its scenario, drafted by the lab from a news report of Wyoming Supreme Court arguments over state trust-land gravel leases, set up a mediation between a mediator, an Assistant Attorney General representing the State Board of Land Commissioners, and the manager of a gravel operator. The names of real people and private organizations in the source were replaced before drafting. The run allowed up to 100 turns of up to 1,000 words each.

Measure	Value
Length	24 turns in 12 min 2 s
How it ended	The moderator judged the scene concluded after the operator's representative withdrew; the assessment found no agreement
Speaker selection	16 answering a direct question, 5 moderator's choice, 2 balancing speaking time; the opening turn by rule
Interventions	Reframe after turn 6, narrow after turn 15, reality test after turn 21
Premature conclusions held back	1, after turn 13
Turns by participant	State's counsel 11, operator 7, mediator 6
Reply length	mean 454 words, maximum 593
Mean model time per call	reply 17.0 s (949 output tokens), ledger 3.7 s, moderator 2.4 s, assessment 35.1 s
Unparseable decisions or ledgers; truncated replies	0; 0

Table 4. The run in numbers.

The mechanisms behaved as specified. Direct questions were routed to the participants asked, all three techniques fired in rotation with the cool-down respected, a conclusion judged after turn 13 was held back because not every participant had yet spoken twice, and the scene ended when one party walked away. A shorter check with three participants, a 150-word limit and a nine-turn cap behaved likewise, with three question-routed turns and one reframe intervention. These are single illustrative runs, not an evaluation.

8 Limitations

- **Question routing can override balance.** Rule 1 precedes the balance rule, and most questions in the run were put to the State's counsel, who took 11 of 24 turns. Capping consecutive question-routed turns for one participant, or applying the balance rule to questions beyond a wider margin, would address this.

- **Interventions can reach the wrong participant.** They are addressed to whoever speaks next, which twice in the run was the neutral mediator, including a reality test that asked the mediator to weigh “your side”. Interventions should be addressed to a party, or phrased for a neutral.
- **The judgements are unvalidated.** Question, stall and conclusion detection come from an 8-billion-parameter model at temperature 0 and have not been compared with human coders.
- **The parameters are heuristic.** The balance margin, the cool-down and the fixed rotation have not been ablated.
- **Outcome fidelity is untested.** Varied-looking simulations can still produce unrealistic outcome distributions [2]; this configuration has not been checked against real negotiation base rates.
- **Scope.** One model on one hardware configuration; the timings do not transfer.

9 Conclusion

Dividing moderation into structured model judgement and code-enforced rules lets a small local model run legal role-plays in which every decision can be inspected afterwards. The live run shows the rules working as designed and points to the next revisions: bounding question routing, and addressing interventions only to parties.

Acknowledgments

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References

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